SPECTRAL EFFECTS OF HEAVY-TAILED VERTEX NOISE IN GEOMETRIC GRAPHS

BEN CARDOEN , JEREMY BUDD , ENRICO AMICO , GHASSAN HAMARNEH , AND FABIAN SPILL

Abstract. We characterize which local matrix structures saturate Weyl's eigenvalue perturbation bound for graph Laplacians under geometrically constrained vertex displacements. Geometric graphs with heavy-tailed vertex noise arise across sensor networks, biological imaging, and spatial omics, yet tractable predictions for noise-induced spectral error remain limited. While Weyl's inequality guarantees eigenvalue continuity, it does not explain how local geometric perturbations propagate to global spectral error, nor how tight these bounds are in structured geometric settings. We study geometric graphs abstracted from biophysical systems, incorporating clearance, planarity, and identifiability constraints that govern physically realizable embeddings. Within this constrained setting, we identify witness motifs, small subgraphs in maximally noise-sensitive geometric configurations, that dominate weighted-degree and graph Laplacian spectral perturbations under tempered power-law vertex displacements. This motif decomposition reduces global spectral sensitivity to a finite catalog of local extremal structures and identifies configurations that attain Weyl-tight bounds. We then lift these constrained-graph results to general straight-line embedded graphs in arbitrary dimension via local repair operations producing a constrained surrogate graph that preserves sensitivity-relevant structure. To quantify noise-induced spectral variation in both strong-oracle (ground truth known) and weak-oracle (observations only) regimes, we introduce stochastic co-spectrality (SC) and the stochastic spectral separation index (S3I), which characterize when observed spectral distances are noise-driven and when noise parameters are separable. Together, these results provide a principled pathway from local geometric noise to global spectral error in graph Laplacian matrices, enabling estimation of spectral fragility from graph structure without exhaustive eigenvalue computation or restrictive distributional assumptions beyond moment bounds.

Key words. spectral graph theory, vertex noise, geometric graphs, heavy-tailed perturbations, graph Laplacian, separability

MSC codes. 05C50, 15A18, 60E07, 62R40

1. Introduction.

Weyl's inequality bounds eigenvalue displacement of a Hermitian matrix under additive perturbation, but does not identify which perturbation structures attain the bound. For graph Laplacians under geometrically constrained vertex noise, we show that a finite catalogue of small subgraph configurations (witness motifs) saturate or nearly saturate this bound (Table 2.1), reducing global spectral sensitivity analysis to local extremal geometry. Concretely, in many physical and biological domains, graphs are not static combinatorial objects but geometric abstractions of spatially embedded systems, constructed from noisy measurements of vertex positions. In super-resolution microscopy, for example, structures are inferred from fluorophore localisations subject to blinking, drift, and finite localisation precision [22]. In imaging-based spatial transcriptomics such as MERFISH, molecules are similarly detected and localised in situ under microscopy noise and occasional outlier errors [5]. In sensor networks, node positions are estimated from imperfect localisation procedures and can exhibit heterogeneous errors and outliers. In such contexts, spectral invariants are frequently employed to compare network topologies or classify structural phenotypes. Because edge weights depend nonlinearly on inter-vertex distance, vertex-position noise induces correlated, non-Gaussian perturbations of edge weights and Laplacians that are not captured by independent edge-noise models. This geometric noise can compress or inflate spectral distances in a way governed by local geometric configurations, motivating structure-aware predictions of spectral error.

While Gaussian models capture small localisation uncertainty, extremal spectral perturbations in geometric graphs are driven by rare moderate displacements. Pure power-law models overemphasise these events and yield infinite-variance regimes incompatible with spectral control. We therefore adopt a calibrated tempered power-law (CTPL) displacement model, that exhibits a power-law regime for moderate jumps with exponential tempering at large radii, ensuring finite moments. Our analysis is application-agnostic and targets the structural mechanisms by which geometric noise propagates from local configurations to Laplacian spectra. We work in a weak-oracle setting in which only perturbed vertex positions are observed (Figure 2.1), rendering all edge weights random even when the underlying graph is fixed. Within this setting, we study both spectral fragility—eigenvalue displacement under noise—and spectral distinguishability—the ability to separate graphs or noise regimes using spectral distances. Although the CTPL model is defined for straight-line embeddings in $\mathbb{R}^d$, all detailed geometric arguments, motif catalogues, and numerical validations in this paper are instantiated in two dimensions. This planar setting captures common spatial networks and

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enforces a finite, explicitly characterisable family of witness motifs via packing constraints; higher-dimensional extensions are discussed only at the level of general formulation.

Classical matrix perturbation theory provides worst-case bounds on eigenvalue and eigenvector changes under additive perturbations [24, 3], but these results are largely distribution-free and do not encode the geometry of vertex noise. Spectral convergence of graph Laplacians has been rigorously analysed for random geometric and point-cloud graphs under static, light-tailed vertex sampling [20, 13, 25], yet these frameworks do not address heavy-tailed geometric perturbations of a fixed underlying graph or link local geometry to global spectral bounds. Related work on network robustness focuses on percolation, edge failures, and degree fluctuations [1, 16], but typically treats vertex positions as fixed or light-tailed latent variables, as in classical random geometric graph models [20].

In this paper, we:

a. introduce a geometric vertex-noise model based on tempered tail (TT) radial displacements, implemented via a CTPL distribution, and show how it induces correlated, non-Gaussian edge-weight perturbations even when the underlying graph is fixed.

b. derive closed-form and asymptotic bounds for spectral displacement by reducing global perturbations to a finite catalogue of extremal witness motifs—small, degree-bounded subgraphs whose geometry certifies worst-case eigenvalue shifts.

c. formalise stochastic co-spectrality and the stochastic spectral separation index as observable summaries that quantify when spectral distances are noise-dominated and when noise parameters cease to be statistically separable.

d. prove that straight-line embedded geometric graphs admit bounded-error reductions via local repair operations, and establish a degree-controlled Frobenius perturbation bound that lifts planar motif-based guarantees to unconstrained embeddings.

2. Methods. A summary of the principal notation introduced in this section is collected in Table 5.1 (Appendix).

2.1. Definitions, notations, and preliminaries. A weighted undirected graph with a straight-line embedding in $\mathbb{R}^d$ is a triple $G = (V, E, W)$, where $V = \{v_1, \dots, v_n\}$ is a finite vertex set, $E \subseteq \{\{v_i, v_j\} \mid i \neq j\}$ is a set of undirected edges, and $W : E \to \mathbb{R}_{>0}$ assigns positive weights to edges. Each vertex v_i is associated with a coordinate $x_i \in \mathbb{R}^d$, inducing a straight-line embedding of G. The weighted adjacency matrix $A \in \mathbb{R}^{n\times n}$ is defined by $A_{ij} = W(\{v_i, v_j\})$ if $\{v_i, v_j\} \in E$ and $A_{ij} = 0$ otherwise, and the weighted degree matrix D is diagonal with entries $D_{ii} = \sum_j A_{ij}$. We use the combinatorial Laplacian $L = D - A$, whose eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ are real. To quantify spectral differences between graphs, we use the one-dimensional Wasserstein–2 distance. Given eigenvalue sets $\boldsymbol{\lambda} = \{\lambda_1, \dots, \lambda_n\}$ and $\boldsymbol{\lambda}' = \{\lambda'_1, \dots, \lambda'_m\}$, define the empirical spectral measures $\mu_G = \frac{1}{n}\sum_{i=1}^n \delta_{\lambda_i}$ and $\mu_{G'} = \frac{1}{m}\sum_{j=1}^m \delta_{\lambda'_j}$. In one dimension, the squared Wasserstein–2 distance admits the explicit form $W_2^2(\mu_G, \mu_{G'}) = \int_0^1 \big(F_G^{-1}(t) - F_{G'}^{-1}(t)\big)^2\, dt$, where F_G^{-1} and $F_{G'}^{-1}$ denote the quantile functions[23]; equivalently, $W_2^2(\mu_G, \mu_{G'}) = \sum_{\ell=1}^L \omega_\ell (x_\ell - y_\ell)^2$ with weights ω_ℓ summing to one and x_ℓ, y_ℓ drawn from the sorted spectra. The spectral distance used throughout the paper is the normalized Wasserstein distance

$$d_{\mathrm{SD}}(G, G') = \frac{1}{\lambda_{\mathrm{ref}}} W_2(\mu_G, \mu_{G'}), \qquad \lambda_{\mathrm{ref}} = \max\big(\|\boldsymbol{\lambda}\|_\infty, \|\boldsymbol{\lambda}'\|_\infty\big). \tag{2.1}$$

Because λ_{ref} depends on both the unperturbed and perturbed spectra, it can be affected by noise where that noise alters the maximal weighted degree. No single normalization scheme is universally optimal: alternatives such as normalizing by the unperturbed $\|\boldsymbol{\lambda}\|_\infty$ alone avoid stochastic denominators but sacrifice symmetry, while fixed scaling constants lose adaptivity to graph size. The choice of λ_{ref} does not affect the theoretical bounds or proofs, which operate on unnormalized spectral shifts; it enters only in the empirical metrics (SC, S3I). The accompanying code permits users to substitute domain-specific normalization as needed.

2.2. Tempered Lévy Distribution. To study heavy tail noise, we need a carefully selected class of distributions. The classical Lévy distribution is a special case of the stable distribution family with stability

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parameter $\alpha = 1/2$ and skewness parameter $\beta = 1$. Stable distributions with $\alpha < 2$ exhibit power-law tails and are often called Lévy α-stable distributions. In this work, we model vertex perturbations using an CTPL distribution, which modifies a power-law tail by exponential decay (Figure 2.2, Panel A). This distribution belongs to the class of tempered stable distributions[21] but avoids the sequential-process connotation of "Lévy process."

2.2.1. Base Lévy–Stable Power-Law Kernel (1D). We model jump magnitudes using the standard one-dimensional Lévy distribution with location μ and scale $c > 0$. Its probability density function on (μ, ∞) is

$$f_{\text{Levy}}(x; \mu, c) = \sqrt{\frac{c}{2\pi}}\,(x-\mu)^{-3/2} \exp\!\left(-\frac{c}{2(x-\mu)}\right), \qquad x > \mu.$$

This is a proper probability density with heavy power-law-like tails and infinite variance in its untempered form. In what follows, this Lévy–stable law serves solely as a canonical power-law kernel: we use it to define a single-step radial displacement model and do not invoke any Lévy flight or process structure.

2.2.2. Calibrated Exponentially Tempered Power Law (CTPL). To ensure the mathematical tractability of spectral observables and the existence of moments, we apply exponential tempering to the Lévy density. Exponentially tempered and truncated Lévy laws are well established as a way to retain Lévy-like jump behaviour while enforcing finite moments and tail control (e.g. [18, 21, 7]). To avoid any ambiguity in terminology and interpretation, although the untempered kernel coincides with the $\alpha = 1/2$ Lévy–stable law, we use CTPL strictly as a single-step radial displacement model and do not assume any Lévy flight or process structure. For parameters $c > 0$, $\lambda \geq 0$, and location μ, define the tempered power-law pdf by

$$\tilde{f}_{\text{temp}}(x; c, \lambda, \mu) = f_{\text{Levy}}(x; \mu, c)\, e^{-\lambda x}, \qquad x > \mu,$$

and let

$$Z(c, \lambda, \mu) = \int_{\mu}^{\infty} f_{\text{Levy}}(x; \mu, c)\, e^{-\lambda x}\, dx < \infty$$

denote the normalising constant. The corresponding calibrated tempered Lévy (CTPL) density is

$$f_{\text{CTPL}}(x; c, \lambda, \mu) = \frac{1}{Z(c, \lambda, \mu)}\, f_{\text{Levy}}(x; \mu, c)\, e^{-\lambda x}, \qquad x > \mu.$$

Exponential tempering ensures the existence of all moments by enforcing rapid decay at infinity, while the Lévy kernel's exponential factor controls the singularity as $x \to \mu^+$; together these imply that $Z(c, \lambda, \mu) < \infty$ and that all moments of f_{CTPL} exist, while the Lévy core retains a heavy-tail-like regime for moderate jump sizes. The parameter c controls the power-law regime, while λ controls the exponential tempering. The 2D isotropic perturbation model is $\boldsymbol{\xi} = R \cdot (\cos\Theta, \sin\Theta)$, where $R \sim \text{CTPL}(c, \lambda)$ and $\Theta \sim \text{Uniform}(0, 2\pi)$ independently. For $r \ll 1/\lambda$, the CTPL density retains a power-law core with exponent $-3/2$ (up to multiplicative constants), so large displacements are substantially more probable than under light-tailed models such as Gaussian noise. For $r \gg 1/\lambda$, exponential tempering dominates and $f_{CTPL}(r) \sim e^{-\lambda r}$, ensuring a well-defined tempered regime. As a consequence of exponential tempering, all moments of the CTPL law exist, guaranteeing that spectral observables such as stochastic co-spectrality (SC) and stochastic spectral separation index (S3I) are well defined. In the calibrated pre-tempering regime, this power-law core yields higher tail probabilities than Gaussian or Gamma alternatives with exponential decay (Figure 2.2, Panel B), inducing heavy-tailed edge-weight noise that cannot be captured by exponentially bounded models. In applications, extremely large jumps are biophysically implausible and also problematic for identifiability. We therefore work with a calibrated and clipped version of the CTPL law. Given a maximum admissible jump radius $r_{\max} > 0$ and a tail tolerance $\delta \in (0, 1)$, we numerically tune the tempering parameter $\lambda = \lambda(c, r_{\max}, \delta)$ so that the untruncated CTPL law satisfies $\Pr[X > r_{\max}] \leq \delta$ for $X \sim \text{CTPL}(c, \lambda, \mu)$ (Figure 2.2, Panel C). Sampling is performed from the corresponding tempered power-law density and deterministically clipped at $r_{\max}$ (Algorithms 1 and 2), so that the law on $[0, r_{\max})$ is unchanged and clipping concentrates at most δ of the total probability mass into an atom at $r_{\max}$; since λ is tuned to enforce $\Pr[X > r_{\max}] \leq \delta$, this atom is controlled by the calibration tolerance. All analytic arguments in this paper are restricted to displacements of magnitude at most $r_{\max}$. Although λ is calibrated only to control the extreme tail beyond $r_{\max}$, varying

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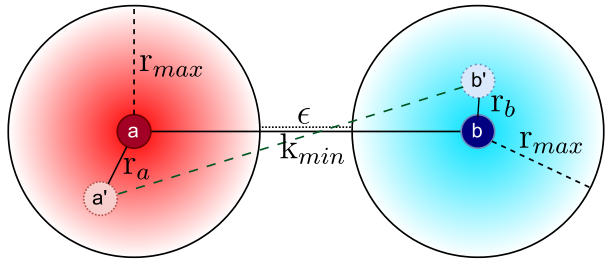

Figure 2.1: Schematic illustration of the vertex noise model (see Section 2.3 for full assumptions A1–A4). Two vertices a and b have latent (noise-free) positions separated by the true edge length k (strong oracle, solid line), where k denotes the ground-truth length of a given edge and satisfies $k \geq k_{\min} = 2r_{\max} + \varepsilon$; the lower bound $k_{\min}$ is a model constraint ensuring that perturbation balls remain ε-separated so that vertices stay identifiable under noise. Each vertex is independently displaced by a radial jump r_a, r_b drawn from a CTPL distribution with uniformly sampled direction, clipped at $r_{\max}$; the perturbed coordinate model is given in Section 2.2.2. The observed edge length (weak oracle, dashed green line) is computed from the perturbed positions a', b'.

the Lévy scale parameter c induces genuinely different behaviour in the core and moderate-tail regimes, so the calibrated CTPL family does not collapse to a single effective noise model.

We distinguish a strong oracle, which has access to the noise-free vertex positions and true edge weights, from a weak oracle, which observes only perturbed vertex positions and hence noisy edge weights (Figure 2.1).

2.3. Geometric Vertex-Noise Graph Model. We consider straight-line embeddings $x : V \to \mathbb{R}^d$ of graphs $G = (V, E)$ under the following geometric and noise assumptions.

(A1) ε-thick embedding (clearance). All non-incident primitives are ε-separated; equivalently, the $\varepsilon/2$-balls around vertices and $\varepsilon/2$-radius tubes around edges are pairwise disjoint. Formally,

$$\begin{aligned} \min\{ \operatorname{dist}(x(u), x(v)),\ \operatorname{dist}(x(u), [x(a), x(b)]), \\ \operatorname{dist}([x(a), x(b)], [x(c), x(d)])\} \ \geq\ \varepsilon, \end{aligned} \tag{2.2}$$

for all distinct $u, v \in V$ and all edges $\{a, b\}, \{c, d\} \in E$ with no shared endpoints. This generalizes the thick embedding constraint studied in geometric graph theory [17, 11, 2] and relates to Federer's notion of positive reach [10] from geometric measure theory.

(A2) Identifiability scale. Let k denote the ground-truth length of a given edge (Figure 2.1), with $k \geq k_{\min} = 2r_{\max} + \varepsilon$; this lower bound ensures that perturbation balls of radius $r_{\max}$ remain ε-separated, so that vertices stay identifiable under noise. More generally, every pair of distinct vertices satisfies $\|x_u - x_v\| \geq k_{\min}$, whether or not $\{u, v\} \in E$; this guarantees that all vertices remain identifiable after perturbation. Neighbours of a vertex lie on (or outside) the k-sphere and must be pairwise ε-separated. Writing

$$\theta = 2 \arcsin\left(\frac{\varepsilon}{2k}\right), \tag{2.3}$$

the maximum degree is bounded by the spherical packing number at angular separation θ [8].

(A3) CTPL vertex noise. Vertices are perturbed independently by an isotropic CTPL radial law as defined in Section 2.2.2. Each displacement has the form $\boldsymbol{\eta} = R\,\mathbf{U}$ (Figure 2.1), where $R \sim \mathrm{CTPL}(c, \lambda, \mu)$ and $\mathbf{U}$ is uniformly distributed on the unit sphere S^{d-1}. The tempering parameter $\lambda = \lambda(c, r_{\max}, \delta)$ is calibrated so that $\Pr[R > r_{\max}] \leq \delta$, and all displacements are truncated to satisfy $R \leq r_{\max}$.

(A4) Weight regularity. Edge weights are bounded and Lipschitz in the inter-vertex distance ρ: there exist $w_{\max}, L_w < \infty$ such that

$$0 \leq w(\rho) \leq w_{\max}, \qquad |w(\rho + \Delta\rho) - w(\rho)| \leq L_w\,|\Delta\rho|. \tag{2.4}$$

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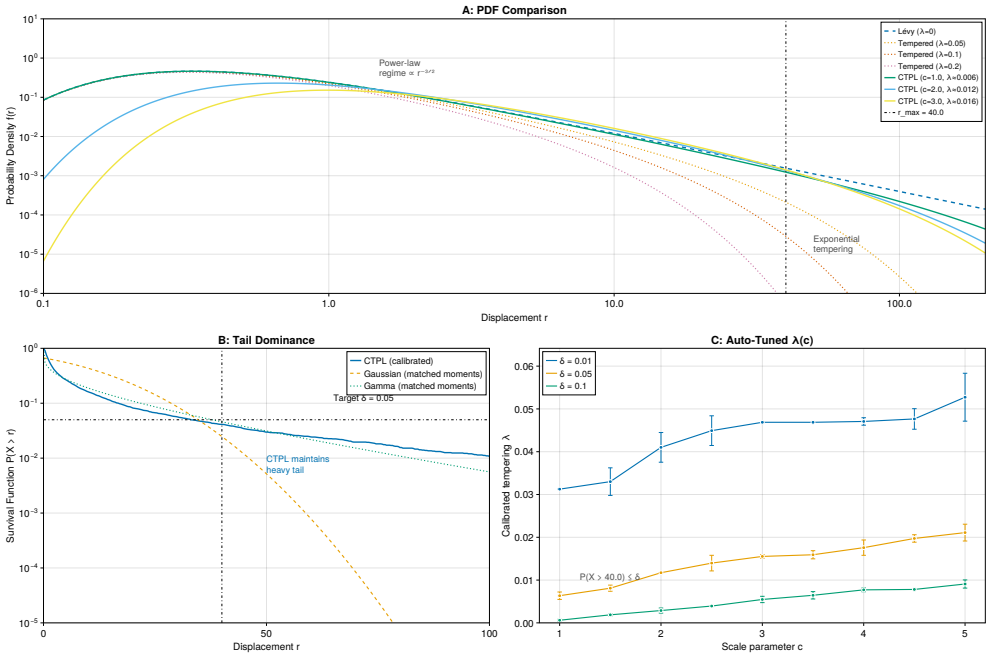

Figure 2.2: From Lévy noise to calibrated tempered power-law (CTPL) noise. **(A)** Probability density functions showing the effect of exponential tempering λ on a Lévy distribution, preserving a power-law regime while suppressing extreme displacements beyond a clipping scale. **(B)** Survival functions demonstrate that the calibrated CTPL distribution retains tail dominance relative to moment-matched Gaussian ($\mu_G = \hat{E}[X_{\mathrm{CTPL}}]$, $\sigma^2 = \widehat{\mathrm{Var}}[X_{\mathrm{CTPL}}]$) and Gamma ($\alpha = \hat{E}[X]^2/\widehat{\mathrm{Var}}[X]$, $\theta = \widehat{\mathrm{Var}}[X]/\hat{E}[X]$) alternatives under a fixed tail probability constraint. **(C)** Empirical auto-tuning of λ as a function of scale parameter c, obtained via rejection sampling to enforce $P(X > r_{\max}) \leq \delta$.

Remark. The extremality results in Section 2.6 additionally require that φ is nondecreasing and convex; all other results (SC, S3I, repair bounds) use only (A1)–(A4) as stated.

Under (A1)–(A3), per-edge changes $|\delta w|$ are bounded, so per-motif perturbations admit finite operator/Frobenius envelopes.

If the ambient metric has finite doubling dimension (every ball of radius r is coverable by N_D balls of radius $r/2$), then packing on the k-sphere obeys

$$\mathrm{Pack}\big(S^{d-1}(k), \varepsilon\big) \;\leq\; C\left(\frac{k}{\varepsilon}\right)^{\log_2 N_D - 1}, \tag{2.5}$$

yielding a finite degree bound [14].

For the remainder of this paper we restrict to displacement metrics $d(x, y) = \|x - y\|$ induced by a norm $\|\cdot\|$ on $\mathbb{R}^d$. Such metrics are convex and monotone nondecreasing in displacement magnitude, the two properties required by the perturbation and extremal arguments in Section 2.6 and Section 2.7. All ℓ^p-norms ($p \geq 1$) are admissible; non-Euclidean choices are angularly preferential and may therefore induce norm-dependent extremal configurations.

2.4. Bounds on Spectral Change in CTPL Graphs. The perturbation of vertices in a graph G leads to a perturbed adjacency matrix A' and a corresponding change in the spectrum. Characterizing this change is central to understanding the graph's fragility. Under light-tailed noise (e.g. Gaussian), extreme displacements are exponentially unlikely; vertex perturbations can be effectively concentrated within one standard deviation, reducing correlated vertex noise to near-independent edge noise for which Weyl's inequality yields tight,

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predictable bounds. Heavy-tailed CTPL noise changes this picture qualitatively. Because the extremal analysis is parametric in the displacement radius, all bounds and motif catalogues can be re-evaluated at any $r \le r_{\max}$. Under CTPL, the power-law core assigns non-negligible probability mass to displacement values well into the tail range—e.g. $0.75\, r_{\max}$—at probabilities far exceeding the calibration cutoff δ. Re-computing the spectral bounds at these sub-maximal but CTPL-probable displacements yields a continuum of practically relevant perturbation estimates, not merely a single worst-case number. Under Gaussian noise, this same range is effectively empty, collapsing the analysis to the near-independent-edge regime. Moreover, even moderate tail values suffice to break the independent-edge-noise reduction, motivating the motif-level analysis that follows. The spectral bounds themselves are deterministic: they hold for any displacement bounded by $r_{\max}$, regardless of the generating distribution. In this section, we bound this change using two complementary tools from matrix analysis: Cauchy interlacing is used to derive deterministic lower bounds from locally perturbed induced subgraphs [12, 6], while Weyl-type perturbation bounds provide corresponding upper bounds on global spectral variation [26, 24, 3].

2.4.1. Lower Bound on Perturbed Eigenvalues via Cauchy Interlacing. We rely on the classical Cauchy interlacing theorem for eigenvalues of principal submatrices [12, 6], which we apply here to derive a lower bound on the spectral effect of localized vertex perturbations.

THEOREM 2.1 (Induced-Subgraph Lower Bound for Perturbed Laplacians). *Let G' be a graph obtained from G by vertex perturbations, and let $H' = G'[V_{H'}]$ be an induced subgraph on a vertex subset $V_{H'} \subseteq V(G')$. Then the largest eigenvalue of the combinatorial Laplacian satisfies*

$$\lambda_n(L_{G'}) \;\ge\; \lambda_{\max}(L_{H'}).$$

Proof. Let $L_{G'}[V_{H'}, V_{H'}]$ denote the principal submatrix of $L_{G'}$ indexed by $V_{H'}$. This matrix decomposes as

$$L_{G'}[V_{H'}, V_{H'}] = L_{H'} + \Delta_{H'}, \qquad \Delta_{H'} := \operatorname{diag}\big(d_{G'}(v) - d_{H'}(v)\big)_{v \in V_{H'}} \succeq 0,$$

where $\Delta_{H'}$ collects degree contributions from edges incident to $V_{H'}$ but not contained in H'. By Cauchy interlacing, the largest eigenvalue of $L_{G'}$ satisfies

$$\lambda_n(L_{G'}) \;\ge\; \mu_m\big(L_{G'}[V_{H'}, V_{H'}]\big),$$

where $\mu_m(\cdot)$ denotes the largest eigenvalue of the principal submatrix. Since $L_{H'} = L_{G'}[V_{H'}, V_{H'}] - \Delta_{H'}$ with $\Delta_{H'} \succeq 0$, monotonicity of eigenvalues under positive semidefinite perturbations yields

$$\mu_m\big(L_{G'}[V_{H'}, V_{H'}]\big) \;\ge\; \lambda_{\max}(L_{H'}).$$

This induced-subgraph bound can be conservative when the witness subgraph omits high-degree connections to the remainder of the graph. In such cases, complementary upper bounds based on operator-norm perturbation theory, such as Weyl-type inequalities and Hoffman–Wielandt bounds [26, 24, 15], may be tighter.

2.4.2. Upper Bound on Eigenvalue Change via Weyl-Type Perturbation Bounds. We now derive a global upper bound on Laplacian eigenvalue displacement under vertex perturbations. This bound follows directly from Weyl's inequality for Hermitian matrix perturbations and complements the induced-subgraph lower bounds of the previous section.

THEOREM 2.2 (Weyl-Type Upper Bound for Laplacian Perturbations). *Let G be a graph with combinatorial Laplacian L_G, and let G' be obtained from G by vertex perturbations, inducing a Laplacian perturbation*

$$E := L_{G'} - L_G.$$

Then, for each $k = 1, \dots, n$,

$$\big|\lambda_k(L_{G'}) - \lambda_k(L_G)\big| \;\le\; \|E\|_2 \;\le\; \|E\|_F. \tag{2.6}$$

Proof. Since L_G and $L_{G'}$ are symmetric, Weyl's inequality for Hermitian matrices [26, 24, 3] implies that

$$\big|\lambda_k(L_{G'}) - \lambda_k(L_G)\big| \le \|E\|_2$$

for all k. The second inequality follows from the standard bound $\|E\|_2 \le \|E\|_F$. ∎

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This bound is global and sharp in principle, but it does not identify which local perturbation patterns realise worst-case spectral displacement. In particular, Weyl-type bounds obscure the role of localized geometric configurations, motivating the motif-based and subgraph-level analyses developed in the following sections.

2.5. The Correspondence between CTPL Vertex and Edge Noise Distributions. Most existing models of noise on graphs act directly on edge weights, typically treating perturbations as independent or weakly dependent across edges, particularly in graph similarity and dynamic network settings [9, 19]. If vertex-level CTPL noise reduced to a simple, tractable edge-noise law, spectral fragility under vertex perturbations could be analysed directly within these frameworks. In general, however, vertex noise induces edge-weight displacements that are nonlinear, correlated, and geometry-dependent. Understanding this distinction is essential, as it motivates the local, motif-based analysis developed in the remainder of this section. Consider a single edge $\{a, b\}$ embedded in $\mathbb{R}^2$ with $\|a - b\| = k$, and let a', b' be obtained by independent vertex perturbations as in Assumption (A3). Write $\eta_a = R_a U_a$ and $\eta_b = R_b U_b$, where $R_a, R_b \sim \mathrm{CTPL}(c, \lambda, \mu)$ are independent radial jump magnitudes and U_a, U_b are independent unit vectors with uniform directions. The noise-perturbed edge length is $k' = \|a' - b'\| = \|(a + \eta_a) - (b + \eta_b)\|$, and the induced edge-length perturbation is $E = k' - k$. In polar coordinates, E is a nonlinear function of the triple (R_a, R_b, ϕ), where ϕ denotes the relative angle between the two perturbations. As a consequence, the induced edge-noise distribution arises as a nontrivial pushforward of the joint vertex-noise law and reflects geometric cancellation and reinforcement effects. Even in this minimal setting, the resulting edge perturbation is neither independent nor expressible in any simple, vertex-agnostic parametric edge-noise model. For multi-edge configurations (Section 2.10), shared vertices introduce strong correlations between adjacent edge perturbations. These dependencies persist under CTPL vertex noise and are clearly visible empirically, even in small geometric motifs (Figure 2.4). As a consequence, vertex noise cannot, in general, be faithfully approximated by independent or identically distributed edge-noise models. This observation renders motif-level analysis essential: local configurations provide the correct scale at which vertex-induced perturbations concentrate and at which worst-case spectral effects can be identified.

2.6. Witness Motifs for Spectral Perturbation. We isolate a bounded family of geometrically admissible subgraphs whose local sensitivity under vertex perturbations yields sharp lower bounds on global spectral change.

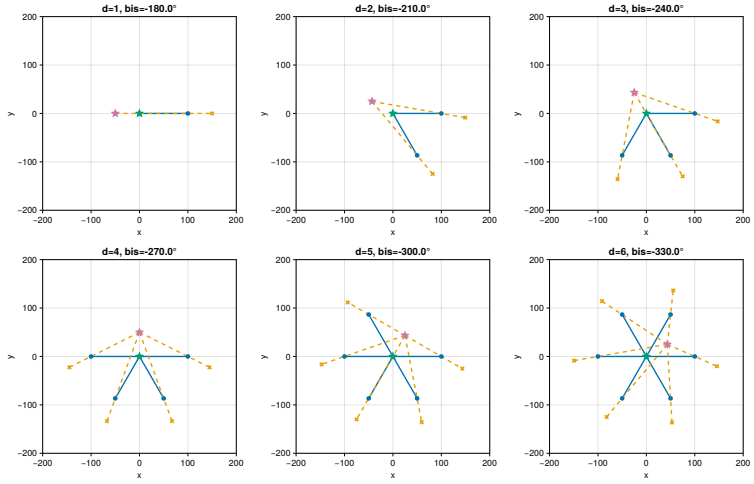

Figure 2.3: Canonical geometric motifs under vertex noise. The six witness motifs considered in this work, shown in two dimensions with a central hub and increasing degree $n = 1$–6. Ground-truth positions are shown in blue (solid edges, filled circles for spokes, green star for hub); perturbed positions in orange (dashed edges, crosses for spokes, purple star for hub). For each motif, the configuration shown corresponds to a maximally sensitive geometric arrangement with respect to vertex noise, aligned along the bisector direction indicated in the panel titles; sensitivity away from this configuration is quantified in Figure 2.6.

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2.6.1. Geometric Packing Bounds on Maximally Sensitive Hub Motifs. We quantify local sensitivity via the relative change in weighted degree at a vertex v under geometric perturbation. Let

$$\mathrm{WD}_G(v) := \sum_{u \in N_G(v)} \varphi(\|x_v - x_u\|),$$

where φ is a differentiable, nondecreasing radial kernel on the relevant distance range, and define for the perturbed graph G'

$$\delta_{\mathrm{rel}}\mathrm{WD}(v) := \frac{\mathrm{WD}_{G'}(v) - \mathrm{WD}_G(v)}{\mathrm{WD}_G(v)}, \qquad \mathrm{WD}_G(v) > 0. \tag{2.7}$$

Under (A1)–(A4), extremal local configurations for $\delta_{\mathrm{rel}}\mathrm{WD}(v)$ arise when many incident edges are simultaneously maximally sensitive to radial displacement while remaining mutually ε-separated. The following two theorems characterise this extremal problem in Euclidean space ($\|\cdot\| = \|\cdot\|_2$): Theorem 2.3 bounds the number of simultaneously sensitive neighbours via a packing argument on $\mathbb{S}^{d-1}$, and Theorem 2.4 locates the maximiser of the weighted-degree displacement within the feasible region. Analogous results hold for other admissible norms, but the packing geometry of the unit ball—and hence the extremal catalogue—is norm-dependent.

Theorem 2.3 (Packing Bound for Maximally Sensitive Hub Motifs). *Let $x_0 \in \mathbb{R}^d$ be a hub vertex with neighbors $x_1, \ldots, x_n$, all at distance $\|x_i - x_0\| = k_{\min}$, under the Euclidean norm and assumptions (A1)–(A4). Then $n \le \tau_d$, where τ_d is the kissing number in dimension d.*

Proof. By (A2), every pair of distinct vertices satisfies $\|x_i - x_j\| \ge k_{\min}$, whether or not they share an edge. Since all neighbors lie at distance $k_{\min}$ from the hub, the law of cosines gives, for $i \ne j$,

$$\|x_i - x_j\|^2 = 2k_{\min}^2(1 - \cos\theta_{ij}),$$

where θ_{ij} is the angle subtended at x_0. The constraint $\|x_i - x_j\| \ge k_{\min}$ yields $\cos\theta_{ij} \le \frac{1}{2}$, i.e. $\theta_{ij} \ge \pi/3$. The directions $u_i = (x_i - x_0)/k_{\min} \in \mathbb{S}^{d-1}$ therefore form a spherical code with minimum angular separation $\pi/3$, whose maximum cardinality is the kissing number τ_d. ∎

In particular, for $d = 2$ we have $\tau_2 = 6$, so the largest geometrically admissible hub–spoke witness has degree six: the hub and its neighbors form equilateral triangles, the tightest configuration permitted by (A2). The extremal regime is $\varepsilon \to 0$, $k = k_{\min} \to 2r_{\max}$, where the noise-to-edge-length ratio—and hence $\delta_{\mathrm{rel}}\mathrm{WD}$—is maximised; for $k > k_{\min}$ the relative perturbation diminishes. The finite catalogue ($n \le 6$) therefore corresponds to the maximally sensitive regime $k = k_{\min}$. The convex boundary argument underlying the extremality theorem below holds in arbitrary dimension d; the restriction to $d = 2$ is made only to obtain the explicit witness catalogue via $\tau_2 = 6$. In higher dimensions, the same construction applies with τ_d replacing τ_2, yielding a larger catalogue whose enumeration we do not pursue here.

2.6.2. Geometric Extremality of Motifs.

Theorem 2.4 (Geometric extremality of hub–spoke weighted-degree displacement). *Let $d = 2$. Let $x_0 \in \mathbb{R}^2$ be a hub and $x_1, \ldots, x_n$ its neighbors with $1 \le n \le \tau_2 = 6$ (Theorem 2.3), all at Euclidean distance $\|x_i - x_0\| = k_{\min}$ and satisfying $\angle x_i x_0 x_j \ge \pi/3$ for $i \ne j$. Let edge weights be $w_i = \varphi(\|x_i - x_0\|)$, where $\varphi : [0, \infty) \to \mathbb{R}$ is differentiable, nondecreasing, and convex. Suppose the hub may move within $B_0 = \{h : \|h - x_0\| \le r_{\max}\}$ and each spoke endpoint within $B_i = \{e : \|e - x_i\| \le r_{\max}\}$. Define the hub weighted-degree change $\Delta W(h, e_1, \ldots, e_n) := \sum_{i=1}^n \big[\varphi(\|e_i - h\|) - \varphi(k_{\min})\big]$. Then there exists an admissible maximizer $(h^\star, e_1^\star, \ldots, e_n^\star)$ of ΔW such that*

(i) $h^\star$ lies on the boundary of B_0, and

(ii) for each i, $e_i^\star$ lies on the boundary of B_i along the ray from $h^\star$ through x_i, pointing away from $h^\star$.

Proof. Fix the spoke endpoints e_i and define $F(h) = \sum_i \varphi(\|e_i - h\|)$. Since φ is convex and nondecreasing and the Euclidean norm $\|\cdot\|$ is convex, each summand $\varphi(\|e_i - h\|)$ is convex in h; hence F is convex. A convex function on a compact convex set attains its maximum on the boundary, so $h^\star \in \partial B_0$; this proves (i).

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Now fix $h = h^\star$ and consider $G_i(e) = \varphi(\|e - h\|)$ for $e \in B_i$. Since φ is nondecreasing, G_i is maximized by choosing e to maximize $\|e - h\|$, which is achieved at the boundary of B_i along the ray from h through x_i; this proves (ii). Replacing any admissible configuration by this boundary configuration cannot decrease ΔW.□

Remark. The angular separation and packing feasibility that bounds admissible motif degree is enforced at the level of motif construction via Assumption (A2) and Theorem 2.3, i.e. on the unperturbed geometry. Theorem 2.4 characterizes where the maximal weighted-degree displacement occurs within a fixed admissible hub–spoke motif, and therefore imposes no additional angular constraint on the perturbed configuration: noise-displaced vertices are not required to preserve pre-noise packing geometry.

In the remainder of this work we use the shorthand G_n for the hub–spoke motifs of degree n constructed above. For such a motif, we write $\delta\mathrm{WD}_{\varepsilon\to 0}(G_n; r) = C_n\, r$ to denote the maximal weighted-degree displacement in the small-noise limit, where C_n is a dimensionless geometric coefficient. The spectral impact of the extremal configurations identified above is assessed empirically in Table 2.1.

For other members of the admissible metric family (§2.3), the structural argument of Theorems 2.3–2.4 carries over, but the packing geometry of the unit ball and the convexity properties of the weight kernel are norm-dependent; consequently, both the degree bound and the boundary-maximiser result must be rederived, yielding a metric-specific extremal catalogue.

2.7. Tractable Estimation via Motif Packing. The Weyl bound in (2.6) controls spectral displacement via the perturbation matrix E, but computing $\|E\|_2$ or $\|E\|_F$ directly requires access to the full noise realization. We obtain a practical and tight estimate by leveraging the analytical extremal behaviour of locally fragile motifs.

Let $\mathcal{M} = \{m_1, \ldots, m_p\}$ be a collection of vertex-disjoint witness motifs embedded in G. Writing the Laplacian perturbation as a sum over motif-restricted contributions,

$$E \;=\; \sum_{i=1}^{p} E_{m_i} \;+\; E_R,$$

where E_{m_i} is the Laplacian perturbation contributed by edges internal to motif m_i—that is, the off-diagonal entries (u, v) for edges $\{u, v\} \subseteq V(m_i)$ and the corresponding diagonal row sums—and E_R collects all remaining contributions, including edges incident to motif vertices but not internal to any single motif. Since the motifs are vertex-disjoint, the matrices $E_{m_1}, \ldots, E_{m_p}$ have pairwise disjoint row/column support, so their Frobenius norms add in quadrature:

$$\sum_{i\neq j} \langle E_{m_i}, E_{m_j}\rangle_F = 0, \qquad \text{hence} \qquad \Big\|\sum_{i=1}^{p} E_{m_i}\Big\|_F^2 = \sum_{i=1}^{p} \|E_{m_i}\|_F^2. \tag{2.8}$$

However, the residual E_R may share row/column support with individual motif blocks (via diagonal degree entries at motif vertices with external edges), so quadrature does not extend to E_R. The triangle inequality yields the always-valid bound

$$\|E\|_F \;\le\; \Big(\sum_{i=1}^{p} \|E_{m_i}\|_F^2\Big)^{1/2} + \|E_R\|_F, \tag{2.9}$$

which retains $\sqrt{p}$ scaling among the motif terms while conservatively bounding the residual contribution.

Combining Equation 2.9 with Weyl's inequality yields the motif-wise spectral bound

$$|\lambda_k(G') - \lambda_k(G)| \;\le\; \|E\|_2 \;\le\; \|E\|_F \;\le\; \Big(\sum_{i=1}^{p} \|E_{m_i}\|_F^2\Big)^{1/2} + \|E_R\|_F,$$

explicitly relating global spectral displacement to local motif contributions. In particular, the motif-only part satisfies

$$\max_i \|E_{m_i}\|_F \;\le\; \Big(\sum_{i=1}^{p} \|E_{m_i}\|_F^2\Big)^{1/2} \;\le\; \sqrt{p}\, \max_i \|E_{m_i}\|_F,$$

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showing that, at the level of this Frobenius-based estimate, repeated occurrences of fragile motifs inflate the motif contribution at most as $\sqrt{p}$. Note that the $\sqrt{p}$ factor is an artifact of the Frobenius norm bound; since the motif blocks have disjoint support, the operator norm satisfies $\| \sum_i E_{m_i} \|_2 = \max_i \|E_{m_i}\|_2$, which is independent of p. The Frobenius route is retained because it extends naturally to the mixed bound with E_R.

To make this bound operational, we replace each $\|E_{m_i}\|_F$ by the analytically derived worst-case value for its motif type. A maximal disjoint set of such motifs can be extracted efficiently using a greedy procedure that prioritizes vertices whose local degree corresponds to high perturbation sensitivity, marking covered vertices to ensure disjointness. The tiling enforces vertex-disjointness only; edges between motifs are permitted, and their perturbative effects are accounted for in the remainder term E_R. The resulting estimate,

$$\Big(\sum_{m_i \in \mathcal{M}} \|E_{m_i,\max}\|_F^2 \Big)^{1/2},$$

provides a computable, data-driven upper bound on the motif contribution to spectral fragility, controlled jointly by motif severity and frequency. Algorithmic details are given in the appendix. Selecting an optimal disjoint motif cover is computationally intractable in general. Indeed, the problem of maximizing total motif coverage under vertex-disjointness constraints reduces to a maximum-weight independent set problem on an associated conflict graph. Since maximum-weight independent set is NP-hard, optimal motif tiling is NP-hard as well; a formal reduction is provided in Appendix Section 5.3. We therefore rely on the greedy heuristic above, which is fast, reproducible, and sufficient for tight practical bounds.

2.8. Spectral Perturbation Under Constraint Repair. This section explains how we extend motif-based perturbation bounds—which are proved under geometric regularity assumptions (e.g., ε-thickness)—to graphs whose embeddings violate those assumptions. The core idea is to repair the embedding into a constrained graph where the motif strategy applies, using local operations that enforce thickness while preserving coarse geometry.

From a geometric graph $G = (V, E)$ embedded in $\mathbb{R}^d$ with straight-line edges, we build a repaired graph $\widetilde{G} = \mathcal{R}(G)$ by applying local edits until all thickness constraints are satisfied. We use the following canonical operations (illustrated in the Algorithms appendix): edge crossing (if two edges cross in their interiors, insert a subdivision vertex at the intersection point and split both edges); vertex collision (if two vertices violate the identifiability/clearance threshold, fuse them into a single vertex, merging incident edges and combining weights according to the model in the Algorithms appendix); and vertex–edge contact (if a vertex lies on the interior of an edge or violates vertex–edge clearance, insert a subdivision vertex at the contact and split the edge).

Each local repair may introduce new constraint violations: splitting an edge can create a segment that passes close to a third vertex, and vertex fusion alters the adjacency structure. The repair therefore proceeds iteratively—via a priority queue in Algorithm 6—until no violations remain. To see that termination is guaranteed, define the discrete potential $\Phi = |V| + C$, where C is the number of geometric constraint violations (edge crossings, vertex–vertex collisions, and vertex–edge proximity violations). Each edit strictly decreases at least one component: vertex fusion reduces $|V|$, and edge subdivision reduces C for the resolved violation without increasing $|V|$. Since Φ is nonnegative and integer-valued, infinite cascading is impossible. We do not claim preservation of global graph distances or spanner properties; the repair serves to enforce the geometric constraints required for subsequent motif analysis. In particular, spectral stability does not follow from metric distortion bounds, and edge weights are not generally additive under subdivision.

Algorithm 6 specifies the repair procedure abstractly; a general-purpose implementation for arbitrary d-dimensional collision resolution is not provided. The experiments in Section 2.11 use a hand-crafted repair tailored to the reference graph, where constraint violations are few and geometrically transparent. In general, automated k-dimensional collision resolution is non-trivial and lies outside the scope of this work: naive implementations risk violating modeling-domain constraints or introducing pathological collapses, particularly when violations are dense or clustered. For example, if all vertices lie within ε of one another, contraction collapses the graph to a single vertex with arbitrarily large spectral deviation; similarly, a high-degree hub with many crossing edges can cause combinatorial blow-up under naive per-crossing subdivision. The perturbation bounds are therefore meaningful in the sparse-violation regime assumed in the experiments.

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The repaired graph (constrained surrogate) $\widetilde{G}$ has vertex set $V_0 \cup V_s$, where V_0 are the original vertices (possibly after fusion) and V_s are deterministic subdivision vertices introduced by repair. By construction, $\widetilde{G}$ satisfies the geometric constraints needed for the motif-tiling bounds in Section 2.7.

To clarify the scope of the subsequent perturbation bound, observe that the total spectral effect of noise on the original graph decomposes as

$$L_{G'} - L_G \;=\; \big(L_{G'} - L_{\widetilde{G}'}\big) \;+\; \big(L_{\widetilde{G}'} - L_{\widetilde{G}}\big) \;+\; \big(L_{\widetilde{G}} - L_G\big).$$

The theorem below controls the middle term—spectral perturbation of the surrogate under vertex noise. The first and third terms represent modeling distortion introduced by repair. Each vertex contraction modifies at most $O(d_{\max})$ rows of the Laplacian, and each edge split modifies $O(1)$ rows, so the total repair distortion scales with the number of edits and the maximum degree. This distortion remains small when repairs are sparse—the regime assumed in the experiments—but may degrade under adversarial configurations; a non-trivial formal bound would require accounting for pathological cases and is not attempted here.

2.8.1. A degree-controlled Laplacian perturbation bound on a repaired graph. We consider i.i.d. CTPL vertex noise acting on the original vertices only. Let $L_{\widetilde{G}}$ be the combinatorial Laplacian of $\widetilde{G}$ and $L_{\widetilde{G}'}$ the Laplacian after perturbing the coordinates of vertices in V_0. Write the Laplacian perturbation as

$$\Delta_{\widetilde{G}} := L_{\widetilde{G}'} - L_{\widetilde{G}}.$$

THEOREM 2.5 (Degree-controlled Frobenius bound and spectral shift control). *Let $H = \widetilde{G}$ be the repaired graph with vertex set $V_0 \cup V_s$, where noise acts only on V_0. Let δw_{uv} denote the change in edge weight on $\{u, v\} \in E(H)$ induced by the noisy coordinates. Define*

$$S(H) := \sum_{\{u,v\}\in E(H)} (\delta w_{uv})^2, \qquad \Delta(H) := \max_{u\in V_0\cup V_s} \deg_H(u).$$

Then the Laplacian perturbation satisfies the Frobenius estimate

$$\|\Delta_H\|_F^2 \;\le\; \big(2 + 2\,\Delta(H)\big)\, S(H). \tag{2.10}$$

Consequently, writing $S_2(L, L') := \big(\sum_{i=1}^n (\lambda_i(L') - \lambda_i(L))^2\big)^{1/2}$ for the ℓ_2 spectral shift, we have

$$S_2(L_H, L_{H'}) \le \|\Delta_H\|_F \;\le\; \sqrt{2 + 2\,\Delta(H)}\,\sqrt{S(H)}. \tag{2.11}$$

Proof. Write $\Delta_H = L_{H'} - L_H$. For a weighted Laplacian, off-diagonal entries satisfy $(L_H)_{uv} = -w_{uv}$ for $\{u, v\} \in E(H)$, hence off-diagonal perturbations contribute

$$\sum_{u\neq v} (\Delta_H)_{uv}^2 = 2 \sum_{\{u,v\}\in E(H)} (\delta w_{uv})^2 = 2\,S(H).$$

Diagonal entries satisfy $(L_H)_{uu} = \sum_{v\sim u} w_{uv}$, so the diagonal perturbation at u is $\sum_{v\sim u} \delta w_{uv}$, and

$$\sum_u (\Delta_H)_{uu}^2 = \sum_{u\in V_0\cup V_s} \Big(\sum_{v\sim u} \delta w_{uv}\Big)^2.$$

For each $u \in V_0 \cup V_s$, Cauchy–Schwarz yields

$$\Big(\sum_{v\sim u} \delta w_{uv}\Big)^2 \le \deg_H(u) \sum_{v\sim u} (\delta w_{uv})^2 \le \Delta(H) \sum_{v\sim u} (\delta w_{uv})^2.$$

Summing over $u \in V_0 \cup V_s$ and re-indexing edge terms gives

$$\sum_{u\in V_0\cup V_s} \Big(\sum_{v\sim u} \delta w_{uv}\Big)^2 \le 2\,\Delta(H) \sum_{\{u,v\}\in E(H)} (\delta w_{uv})^2 = 2\,\Delta(H)\,S(H).$$

Combining diagonal and off-diagonal contributions yields (2.10). The spectral-shift inequality (2.11) follows from the Hoffman–Wielandt theorem [15]. ∎

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2.9. Stochastic spectral metrics for observability and separability. To quantify how vertex noise affects spectral measurements, we do not introduce new spectral distances. Instead, we formalize moment-based summaries of existing spectral distances in order to assess (i) whether spectral differences are observable under noise, and (ii) whether different noise regimes can be distinguished from spectral data alone. Fix a graph G, a noise model $\mathcal{N}$, and a spectral distance d_{SD}. Let $D := d_{SD}\big(G, \mathcal{N}(G)\big)$ denote the induced spectral-distance random variable.

2.9.1. Stochastic co-spectrality (SC). We define the stochastic co-spectrality moments as the raw moments of D:

$$\mathrm{SC}_k(G;\mathcal{N}) := \mathbb{E}[D^k], \qquad k = 1, 2, \ldots$$

whenever they exist. In particular,

$$\mu_D := \mathrm{SC}_1(G;\mathcal{N}), \qquad \sigma_D^2 := \mathrm{SC}_2(G;\mathcal{N}) - \mathrm{SC}_1(G;\mathcal{N})^2.$$

These quantities summarize the expected spectral displacement induced by noise and its variability. For untempered heavy-tailed noise models, such moments need not exist; under the calibrated CTPL model, all moments required below are finite. A central use of SC is to assess noise observability. Let $D^\star := d_{SD}\big(\mathcal{N}(G), \mathcal{N}(G')\big)$ be an observed spectral distance between two noisy graphs. Using Cantelli's inequality [4], if μ_D and σ_D^2 exist then $\Pr\Big[D \geq \mu_D + \sigma_D\sqrt{\frac{1-\alpha}{\alpha}}\Big] \leq \alpha$. We therefore say that $D^\star$ is noise-observable at level α if $\frac{D^\star - \mu_D}{\sigma_D} > \sqrt{\frac{1-\alpha}{\alpha}}$.

2.9.2. Observable SC from noisy data (weak-oracle setting). When the noise-free graph G is unavailable, SC can still be estimated from paired noisy observations. Let $G_i, G_j \sim \mathcal{N}(G)$ be independent perturbations. Define

$$\mathrm{SC}_{\mathrm{obs}}(G;\mathcal{N}) := \mathbb{E}[d_{SD}(G_i, G_j)]\,.$$

By the triangle inequality,

$$\mathbb{E}[|D_i - D_j|] \;\leq\; \mathrm{SC}_{\mathrm{obs}}(G;\mathcal{N}) \;\leq\; 2\,\mathbb{E}[D],$$

so $\mathrm{SC}_{\mathrm{obs}}$ provides an observable proxy for spectral compactness under noise, with finite variance whenever μ_D and σ_D^2 exist.

2.9.3. Stochastic spectral separation index (S3I). To compare two noise models $\mathcal{N}_a$ and $\mathcal{N}_b$ acting on the same graph G, let

$$D^{(a)} := d_{SD}\big(G, \mathcal{N}_a(G)\big), \qquad D^{(b)} := d_{SD}\big(G, \mathcal{N}_b(G)\big),$$

with means μ_a, μ_b and variances σ_a^2, σ_b^2. We define the stochastic spectral separation index as

$$\mathrm{S3I}(G;\mathcal{N}_a,\mathcal{N}_b) := \frac{|\mu_a - \mu_b|}{\sqrt{(\sigma_a^2 + \sigma_b^2)/2}}.$$

This quantity is an effect-size-style summary, analogous to Cohen's d, measuring how distinguishable the average spectral impacts of two noise regimes are relative to their pooled variability. In practice, S3I is estimated from empirical samples (either strong-oracle or weak-oracle), and is used descriptively rather than as a formal separability guarantee. In empirical settings, both SC and S3I are estimated from finite samples of noisy realizations. To assess uncertainty we compute S3I independently over N replicated Monte Carlo runs, each drawing fresh noise realisations, and report the $(1-\alpha)$ percentile interval of the resulting S3I distribution. If this interval excludes zero, the two noise regimes are deemed empirically separable under the $(1-\alpha)$ percentile criterion. This procedure is used throughout the experimental results to assess robustness of spectral distinguishability.

2.10. Extent of vertex–edge noise distribution divergence. Figure 2.4 illustrates a fundamental distinction between vertex- and edge-based noise models using an asymmetric three-spoke hub motif ($k = 100$, $r_{\max} = 50$, spokes at $0°$, $60°$, $210°$; 500 noise samples per condition, calibration tolerance $\delta = 0.05$). Even for this minimal configuration, independent vertex perturbations induce correlated and geometry-dependent edge-weight fluctuations. This confirms that vertex noise cannot, in general, be reduced to independent or identically distributed edge noise, motivating the motif-level analysis developed in the Methods section.

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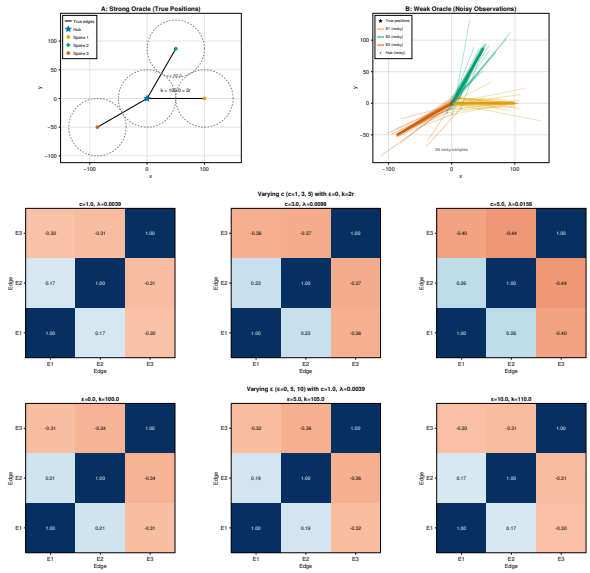

Figure 2.4: Vertex noise induces correlated edge perturbations. (A) Strong oracle: an asymmetric three-spoke hub motif at $k = 2r_{\max}$ with ground-truth positions (blue star = hub, colored circles = spokes) and perturbation balls (gray dashed). (B) Weak oracle: 50 noisy realisations overlaid; faded black lines and markers show ground-truth positions for reference, coloured semi-transparent lines show perturbed edges. Bottom rows: empirical edge-length correlation matrices (500 samples each). Row 2 varies the scale parameter c at fixed $\varepsilon = 0$ (with λ recalibrated per c); Row 3 varies ε at fixed $c = 1$. Off-diagonal correlations are consistently nonzero, confirming that vertex noise induces structured edge dependencies even in minimal configurations.

2.11. Empirical spectral signatures of CTPL noise. We evaluate S3I on the reference graph of Figure 5.1 (see Appendix) ($k = 100$, inter-motif spacing 300; 28 vertices, 27 edges), using calibrated CTPL noise with tolerance $\delta = 0.05$. Each comparison comprises 100 independent runs of 200 noise samples, with 95% percentile intervals reported. Panel A compares $c_1 = 1$ against $c_2 \in \{2, 3, 4, 5\}$ at fixed $r_{\max} = 40$; Panel B compares $r_{\max} = 10$ against $r_{\max} \in \{20, 30, 40, 50\}$ at fixed $c = 1$; Panel C compares the heterogeneous reference graph against a uniform (6× degree-4) variant at $c \in \{1, \ldots, 5\}$; Panel D compares an unconstrained graph (hub removed, crossing edges) against its ε-thick repaired counterpart at $c \in \{1, \ldots, 5\}$.

Figure 2.5 demonstrates that S3I detects multiple, qualitatively distinct sources of spectral variation. Varying the CTPL scale parameter c (Panel A) or the maximal jump radius $r_{\max}$ (Panel B) produces clear spectral separation, with all confidence intervals well above zero, indicating sensitivity to noise parameters alone. Changing motif composition while holding global structure fixed (Panel C) yields a weaker but detectable separation: S3I is consistently positive, though the lower confidence bound approaches zero at low noise, indicating that separability diminishes as perturbation magnitude decreases. Similarly, enforcing ε-thickness via local repair (Panel D) produces a bounded spectral shift consistent with the degree-controlled perturbation framework of Section 2.7, with lower bounds close to zero at low c. Each panel is a pairwise comparison at a given operating point; no cross-comparison between operating points (e.g. across c values) is tested.

Together, these experiments confirm that S3I provides a practical, observable summary of how noise parameters, geometric constraints, and local graph structure jointly shape spectral variability under vertex noise.

2.12. Motif geometry sensitivity. Figure 2.6 empirically validates the geometric extremality assumptions underlying the motif analysis. For each degree $n = 1, \ldots, 6$, hub–spoke motifs are constructed at $k = 100$ with $r = 50$; the hub displacement angle is swept over 121 grid points in $[-60°, 60°]$ relative to the bisector, with 50 random spoke-angle perturbations per grid point. For all degrees, maximal sensitivity is attained at the canonical configurations identified analytically, and angular deviations strictly reduce the induced

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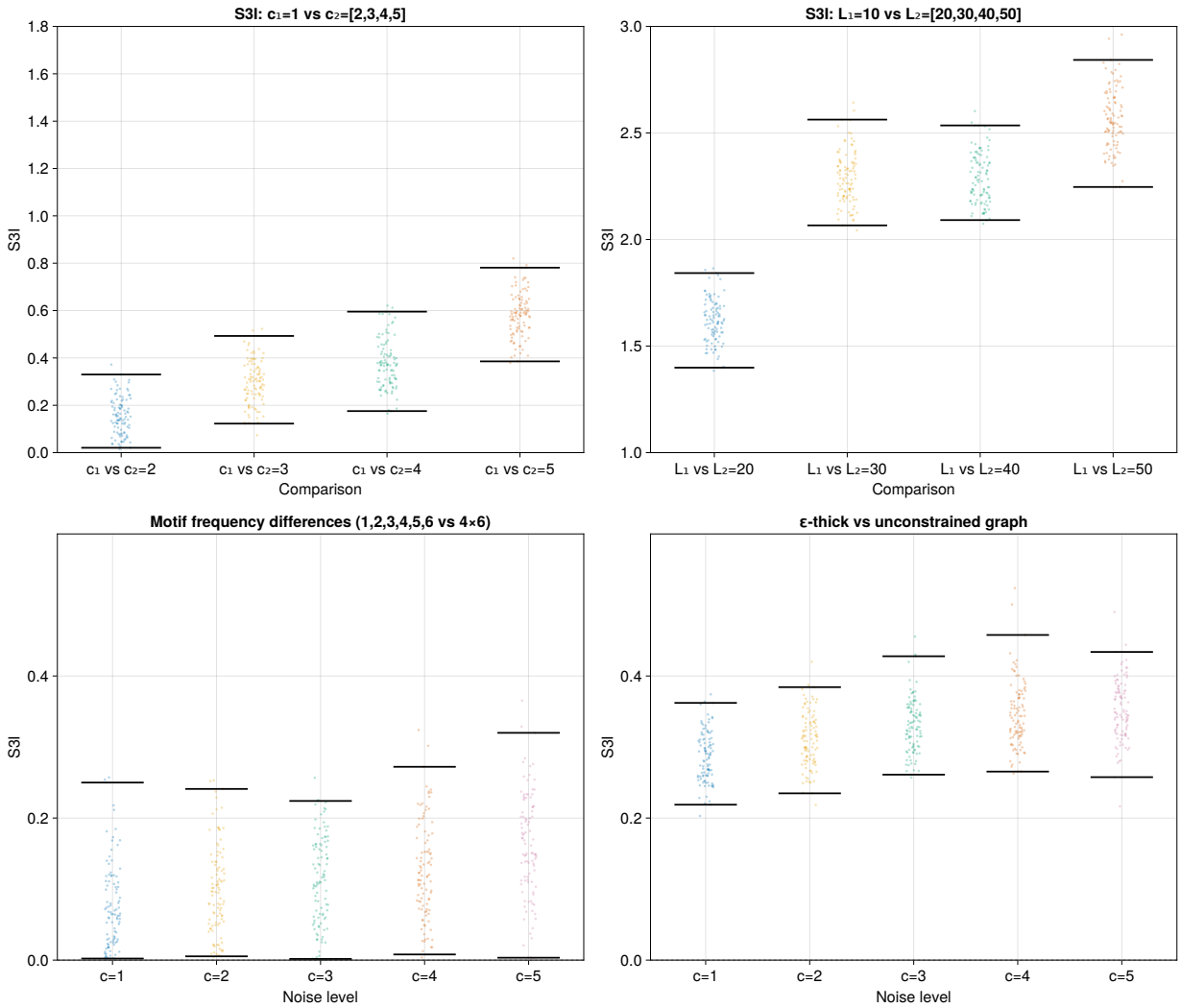

Figure 2.5: Motif-informed graph-level sensitivity and repair effects. Each panel reports S3I values from 100 independent Monte Carlo runs (200 noise samples each), with 95% percentile confidence intervals (black bars). A checkmark indicates that the interval excludes zero, i.e. S3I is consistently positive across runs. (A) Separation induced by changes in the scale parameter c at fixed $r_{\max}$. (B) Separation induced by changing $r_{\max}$ at fixed c. (C) Separation between graphs with different motif compositions (heterogeneous 1–6 versus uniform 4×6) under identical noise. (D) Separation between an unconstrained graph and its ε-thick repaired counterpart. Note that each comparison is a pairwise test at a single operating point; S3I values should not be compared across operating points without further testing.

weighted-degree change. This confirms that the witness motifs used throughout the paper are not artefacts of a particular alignment but represent genuine local maxima under the admissible geometric constraints.

2.13. Spectral tightness of witness motifs. Table 2.1 reports maximal weighted-degree perturbations for each witness motif in the extremal regime ($k = 2r$, $\varepsilon \to 0$, $r = 1$), together with the corresponding observed maximum Laplacian eigenvalue shift and Weyl upper bound ($\|L' - L\|_2$); the ratio column gives the tightness of the Weyl bound (shift/bound). Low-degree motifs (degrees 1 and 2) exhibit numerical Weyl tightness (ratio $= 1$), while higher-degree motifs show increasing slack, indicating that spectral misalignment—rather than noise magnitude alone—limits tightness. These results justify the motif ranking used in the packing-based fragility bounds and support the use of low-degree motifs as sharp spectral witnesses in practice.

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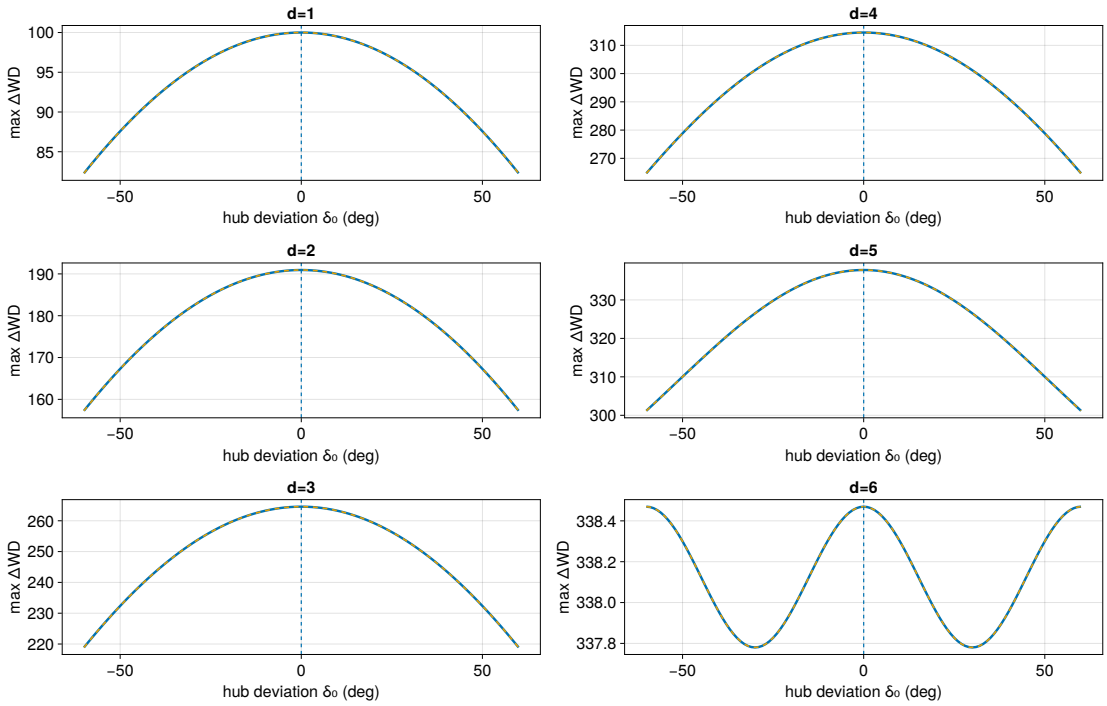

Figure 2.6: Sensitivity of witness motifs to angular perturbations. For each motif degree $n = 1, \ldots, 6$, the maximum change in weighted degree ΔWD is shown as a function of the hub displacement angle relative to the optimal bisector axis of the unoccupied arc. For $n \leq 5$, the extremum is unique; for $n = 6$, hexagonal symmetry induces multiple equivalent maxima. The admissible angular range is determined by the ε-thickness constraint.

Table 2.1: Maximum relative change in weighted degree for maximally affected subgraphs under CTPL noise, together with observed spectral displacement and Weyl bounds. Here r denotes $r_{\max}$ (the extremal regime sets $k = 2r_{\max}$, $\varepsilon \to 0$).

Degree	δ WD $\varepsilon \to 0$	Actual Shift	Weyl Limit	Ratio
1	$2r$	4.0000	4.0000	1.0000
2	$3.82r$	5.7279	5.7279	1.0000
3	$5.29r$	7.0653	7.0768	0.9984
4	$6.29r$	7.9030	7.9523	0.9938
5	$6.76r$	8.2030	8.3373	0.9839
6	$6.77r$	8.0702	8.3460	0.9670

3. Discussion and Conclusions. This work formalises vertex noise in geometric graphs through calibrated tempered power-law (CTPL) perturbations and introduces two complementary stochastic spectral quantities: stochastic co-spectrality (SC), which captures the distributional footprint of spectral distances under noise, and the stochastic spectral separation index (S3I), which summarises the distinguishability of noise regimes from spectral data alone. Together with motif-based perturbation analysis, these quantities provide a principled bridge between local geometric noise mechanisms and global spectral effects.

A central insight of the paper is that vertex noise induces structured, correlated, and geometry-dependent edge perturbations that cannot, in general, be reduced to independent edge-noise models. This motivates the use of local geometric witness motifs as the correct scale at which worst-case and aggregate spectral effects

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concentrate. By combining sharp local extremal analysis with Weyl-type and Frobenius bounds, we obtain tractable estimates of global spectral fragility that depend explicitly on motif frequency and degree structure rather than on opaque global constants. The resulting framework cleanly separates where fragility originates (local geometry) from how it propagates (spectral perturbation theory).

From an algorithmic perspective, the greedy motif-tiling strategy used here is deliberately simple. The associated optimal tiling problem is NP-hard, so exact solutions are infeasible at scale; nevertheless, the empirical results indicate that even coarse tilings capture the dominant contributors to spectral instability. Designing faster or approximation-guaranteed motif extraction schemes is a natural direction for future work. The theoretical framework—Weyl-type bounds, Frobenius decomposition, and the motif-packing strategy—holds in any dimension under assumptions (A1)–(A4), but the specific witness catalogue and empirical tightness constants (Table 2.1) are derived and validated in two dimensions. Extending the motif catalogue and sharp constants to higher-dimensional settings, where spatial saturation constraints tighten and the family of extremal configurations diversifies, will likely require new structural insights.

While our analysis is restricted to norm-induced distances, a significant part of the community works with similarity-based metrics, where proximity increases, rather than decreases the weight of an edge. Due to space limitations we cannot fully explore this direction here. Extending the present bounds to similarity kernels is non-trivial: decreasing or near-singular kernels (e.g. inverse-distance) amplify small post-noise separations, breaking the uniform perturbation envelopes used above unless additional post-noise separation or kernel regularization is imposed. The obstruction is therefore analytic rather than computational. We note that our accompanying open-source packages (Section 5.2) support arbitrary kernels, so practitioners can empirically assess the impact of their chosen kernel on spectral fragility and separability in their own settings.

Beyond methodology, the framework has direct relevance for spatial networks arising in imaging and biological reconstruction, such as mitochondrial and endoplasmic reticulum networks, filament skeletons, and single-molecule localisation graphs. In these settings, perturbations are naturally vertex-local, heavy-tailed, and geometry-constrained. The quantities introduced here—SC and S3I—offer interpretable summaries of spectral robustness and separability that remain meaningful even when the underlying noise distributions are non-Gaussian and only weak-oracle observations are available. In particular, estimating the noise-induced spectral envelope provides a computable noise floor that can replace domain-specific simulations or edge-independent approximations, enabling practitioners to delineate genuine structural change from localisation-induced spectral artefacts and thereby reduce both false positives and overly conservative false-negative thresholds. The extension to unconstrained embeddings via repair ensures this capability is not limited to biophysically motivated networks, and the accompanying open-source packages provide domain experts with ready-to-use tooling for spectral noise quantification on their own geometric configurations.

A longer-term implication is the possibility of inversion: given noisy spectral observations and partial geometric priors, can one constrain or recover parameters of the underlying vertex-noise process using motif-level approximations and stochastic spectral summaries? Addressing such inverse problems, alongside scalable motif detection in higher dimensions, would further connect the present theory to data-driven applications. Taken together, the results of this work establish motif-aware stochastic spectral analysis as a viable and flexible lens for studying vertex noise in geometric graphs.

4. References.

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